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DC Practical 8
Problem Statement: Implement DEAP (Distributed Evolutionary Algorithms) using Python.

Code:

```
import random
from deap import base, creator, tools, algorithms
import multiprocessing

# Fitness function (minimize)
def eval_func(individual):
    return sum(x**2 for x in individual),

# DEAP Setup
creator.create("FitnessMin", base.Fitness, weights=(-1.0,))
creator.create("Individual", list, fitness=creator.FitnessMin)

toolbox = base.Toolbox()

toolbox.register("attr_float", random.uniform, -5.0, 5.0)
toolbox.register("individual", tools.initRepeat, creator.Individual, toolbox.attr_float, n=3)
toolbox.register("population", tools.initRepeat, list, toolbox.individual)

toolbox.register("evaluate", eval_func)
toolbox.register("mate", tools.cxBlend, alpha=0.5)
toolbox.register("mutate", tools.mutGaussian, mu=0, sigma=1, indpb=0.2)
toolbox.register("select", tools.selTournament, tournsize=3)

# PARALLELIZATION (IMPORTANT)
pool = multiprocessing.Pool()
toolbox.register("map", pool.map)

# Create population
population = toolbox.population(n=50)

# Evaluate initial population
fitnesses = list(toolbox.map(toolbox.evaluate, population))
for ind, fit in zip(population, fitnesses):
    ind.fitness.values = fit

# GA loop
generations = 20

for gen in range(generations):
    offspring = algorithms.varAnd(population, toolbox, cxpb=0.5, mutpb=0.1)
```

```
    fitnesses = list(toolbox.map(toolbox.evaluate, offspring))
    for ind, fit in zip(offspring, fitnesses):
        ind.fitness.values = fit

    population = toolbox.select(offspring, k=len(population))

    best = tools.selBest(population, k=1)[0]
    print(f"Generation {gen}: Best Fitness = {best.fitness.values[0]:.4f}")

# Final result
best_ind = tools.selBest(population, k=1)[0]
print("\nBest Individual:", best_ind)
print("Best Fitness:", best_ind.fitness.values[0])

pool.close()
pool.join()
```

Output:

```
vboxuser@Ubuntu-24: ~/Desktop/CL3/DC8
vboxuser@Ubuntu-24:~/Desktop/CL3/DC8$ python3 DC8.py
Generation 0: Best Fitness = 2.4443
Generation 1: Best Fitness = 2.1525
Generation 2: Best Fitness = 1.0013
Generation 3: Best Fitness = 0.6111
Generation 4: Best Fitness = 0.4594
Generation 5: Best Fitness = 0.2825
Generation 6: Best Fitness = 0.1292
Generation 7: Best Fitness = 0.0662
Generation 8: Best Fitness = 0.0662
Generation 9: Best Fitness = 0.0662
Generation 10: Best Fitness = 0.0265
Generation 11: Best Fitness = 0.0260
Generation 12: Best Fitness = 0.0151
Generation 13: Best Fitness = 0.0151
Generation 14: Best Fitness = 0.0079
Generation 15: Best Fitness = 0.0067
Generation 16: Best Fitness = 0.0040
Generation 17: Best Fitness = 0.0030
Generation 18: Best Fitness = 0.0030
Generation 19: Best Fitness = 0.0027

Best Individual: [0.017707991044473962, -0.004309355082239163, 0.048516389301782245]
Best Fitness: 0.002685983518938081
vboxuser@Ubuntu-24:~/Desktop/CL3/DC8$
```